

Exam 1
ECON 306 — Intermediate Microeconomic Theory
Exam 1, Summer Session I 2026

Instructions

- Write the answer to questions on a separate sheet of paper.
- DO NOT WRITE YOUR ANSWERS ON A TABLET OR IPAD. USE OF THOSE DEVICES IS STRICTLY PROHIBITED DURING THE EXAM.
- Each question is 1 point unless otherwise marked.
- Answers without appropriate work may not receive full credit so please show your work!
- Partial credit may be given but it is up to my discretion
- The last page contains a formula sheet.
- Unless otherwise specified, assume preferences are well behaved.
- Round answers if needed to the nearest 1/100th (0.01).
- A calculator may be used.

Scenario 1

The supply and demand for widgets is given by:

$$Q_D = 30 - 0.5p, \quad Q_S = 2p - 50.$$

1. (2 points) At the equilibrium price and quantity, find the price elasticity of demand (E_D) and the price elasticity of supply (E_S).
2. (2 points) A per-unit tax of $t = 10$ is imposed on sellers. What is the equilibrium price that sellers receive, the price that buyers pay and the quantity with the tax?
3. (2 points) Calculate the tax incidence (\$ amount per unit bought/sold) — i.e., the buyer's burden and the seller's burden.

Scenario 2

Sandy from Sandusky has the utility function

$$U(q_1, q_2) = q_1^{0.5} q_2^{0.5}.$$

Sandy has an income of $Y = 2000$ and the prices are $p_1 = 5$ and $p_2 = 20$.

4. Write the consumer's budget constraint.
5. (2 points) Find the consumer's optimal consumption bundle.

Suppose the prices rise to $p_1 = 10$ and $p_2 = 30$.

6. How much income would Sandy now require to be able to afford the bundle from the previous question?

7. (2 points) Jeanne, a caffeine-deprived student, needs to wake up. Suppose her preferences can be described by only caring about the total amount of caffeine consumed — she wants to consume the most she can afford. Tea (q_1) has 40 mg of caffeine per cup, and coffee (q_2) has 60 mg per cup. The price of tea is \$1 per cup and the price of coffee is \$4 per cup. She has \$20 of income to spend. What is her utility function, and what is the name of this kind of preference?
8. (2 points) Continuing the previous question, what is her optimal consumption bundle?
9. (2 points) Suppose Frank, another caffeine-deprived student, needs to wake up. He always makes his morning coffee with 1 cup of black coffee (q_1) and 4 packets of hazelnut creamer (q_2). He wants to make as many coffee drinks as he can afford, but refuses to drink a cup of coffee with less or more than 4 packs of creamer per cup; any extra creamer beyond that is thrown away, any coffee without creamer is thrown away. Coffee costs \$4 per cup, and hazelnut creamer costs \$0.25 per packet. He has \$20 to spend and wants to drink as many perfect cups of coffee as possible. What is a utility function representing these preferences, and what is the name of this kind of preference?
10. (2 points) Continuing the previous question, what is Frank's optimal consumption bundle?

Scenario 3

Suppose a firm has the production function

$$Q = 15 K L^2 - 3 L^3.$$

Capital is fixed at $K = 18$.

11. At what quantity of labor does diminishing marginal returns to labor begin?
12. What is the maximum output this firm can produce with the current amount of capital?

Scenario 4

The daily production of widgets requires workers (L) and machines (K). The production function is

$$Q = 5 L^{0.25} K^{0.25},$$

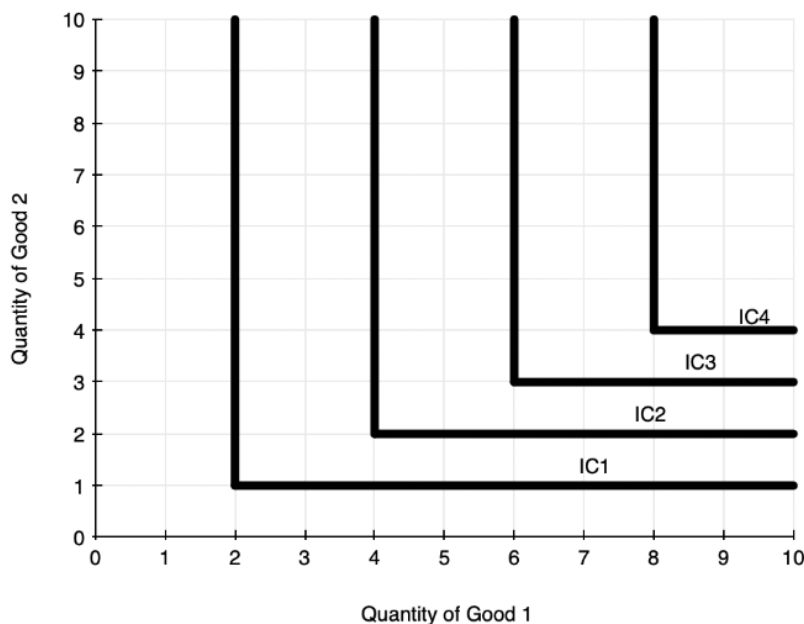
where Q is the quantity of widgets. The wage for workers is $w = 16$ and the rental rate of machines is $r = 400$. Initially, the quantity of machines is fixed at $K = 800$. The firm wants to produce 100 widgets.

13. What are the returns to scale of this production function?
14. In the short run, how many workers would the firm need to employ to produce 100 widgets?

15. (2 points) Now suppose the firm is looking to the long run. If the firm plans to continue producing 100 widgets, how much labor and capital will it employ?
16. How much does it cost to produce 100 widgets in the long run?

Multiple Choice Questions

17. A curve that represents all combinations of market baskets that provide the same level of utility to a consumer is called:
- A. a budget line.
 - B. a utility function
 - C. an indifference curve.
 - D. a demand curve.
18. Consider a consumer who can buy only two goods: Pizza (q_1) and Pepperoni (q_2) with q_1 on the x-axis. The consumer prefers more pizza to less but is indifferent between how much pepperoni consumed. What will be the shape of the consumer's indifference curves?
- A. Downward sloping
 - B. L-shaped
 - C. Linear and downward sloping
 - D. Vertical
19. Which of the following utility functions represents the preferences for the indifference curves shown below?



- A. $U(q_1, q_2) = \min\{2q_1, q_2\}$
 - B. $U(q_1, q_2) = \min\{q_1, q_2\}$
 - C. $U(q_1, q_2) = \min\{q_1, 2q_2\}$
 - D. $U(q_1, q_2) = q_1 + 2q_2$
20. A firm's production function is $Q = 10 L^{0.4} K^{0.4}$. This production function exhibits:
- A. increasing returns to scale
 - B. constant returns to scale
 - C. decreasing returns to scale
 - D. returns that cannot be determined without input prices
21. Which of the following best describes the law of diminishing marginal returns?
- A. Output falls when a firm becomes too large
 - B. Holding other inputs fixed, the marginal product of a variable input eventually decreases as more of it is used
 - C. Doubling all inputs less than doubles output
 - D. Average fixed cost rises as output increases
22. A utility-maximizing consumer who buys positive amounts of both goods 1 and 2 will choose a bundle where:
- A. the marginal utility of good 1 equals the marginal utility of good 2
 - B. total utility from good 1 equals total utility from good 2
 - C. the marginal rate of substitution equals the price ratio p_1/p_2
 - D. the indifference curve crosses the budget line
23. In general, if the burden of a tax falls more on sellers than on buyers, this implies:
- A. Demand is more price elastic than supply
 - B. Supply is more price elastic than demand
 - C. The tax was collected from the sellers
 - D. The tax is unusually high relative to equilibrium price
24. An upward sloping indifference curve violates which of the following axioms of consumer choice?
- A. transitivity.
 - B. complete preferences
 - C. convexity
 - D. more is preferred to less

25. A firm is producing a level of output at an input combination where its isoquant is steeper than its isocost line meaning the marginal rate of technical substitution exceeds the input-price ratio ($MRTS > w/r$). To produce the *same* output at lower cost, the firm should:
- A. use more capital and less labor
 - B. use more labor and less capital
 - C. increase its use of both labor and capital
 - D. make no change, since it is already minimizing cost

Formula Sheet

Elasticities

$$E = \frac{\partial Q}{\partial p} \cdot \frac{p}{Q}, \quad E_D = \frac{\partial Q_D}{\partial p} \cdot \frac{p}{Q_D}, \quad E_S = \frac{\partial Q_S}{\partial p} \cdot \frac{p}{Q_S}, \quad E_Y = \frac{\partial Q_D}{\partial Y} \cdot \frac{Y}{Q_D}$$

MRS and MRT

$$MRS = -\frac{MU_1}{MU_2}, \quad \text{where } MU_i = \frac{\partial U}{\partial q_i}; \quad MRT = -\frac{p_1}{p_2}$$

Perfect Substitutes

$$U(q_1, q_2) = aq_1 + bq_2$$

Perfect Complements

$$U(q_1, q_2) = \min\{aq_1, bq_2\}$$

Cobb–Douglas

$$U(q_1, q_2) = q_1^a q_2^b, \quad MRS = -\frac{a q_2}{b q_1}$$

Production

$$Q = f(K, L), \quad MP_L = \frac{\partial Q}{\partial L}, \quad MP_K = \frac{\partial Q}{\partial K}, \quad AP_L = \frac{Q}{L}$$

$$MRTS = -\frac{MP_L}{MP_K} \text{ (slope of isoquant),} \quad \text{slope of isocost} = -\frac{w}{r}$$

Examples of Production Functions

$$\text{Perfect Substitutes: } f(K, L) = aK + bL, \quad a > 0, b > 0$$

$$\text{Fixed Proportions: } f(K, L) = c \min\{aK, bL\}, \quad a, b, c > 0$$

$$\text{Cobb–Douglas: } f(K, L) = A K^a L^b, \quad a, b, A > 0$$

$$\text{Cobb–Douglas } MRTS = -\frac{MP_L}{MP_K} = -\frac{b K}{a L}$$

Cost of Production

$$\text{Total cost of inputs} = wL + rK, \quad AC(Q) = \frac{C(Q)}{Q}, \quad MC(Q) = C'(Q)$$

$$C(Q) = VC(Q) + FC, \quad AVC(Q) = \frac{VC(Q)}{Q}, \quad AFC(Q) = \frac{FC}{Q}$$

Production and Cost Relationship

$$MC = \frac{w}{MP_L}, \quad AVC = \frac{w}{AP_L}$$